

INTERNAL

Author: Cerberus Research

Generated: 2026-08-05 08:29 UTC

N 3	EDGE +2.58%	P 0.207	HIT-RATE 66.7%
CLASSIFICATION internal	AUTHOR Cerberus Research		
RUN ID 20260805_092910	GIT COMMIT aa4576bd		
DURATION 0.5s	DATASETS duckdb:crypto_binance_oi_data		
PARAMETER · SYMBOL BTCUSDT	PARAMETER · BENCHMARK BTC-USD		
PARAMETER · LOOKBACK 20	PARAMETER · HORIZON 7		
PARAMETER · Z_THRESHOLD 2	PARAMETER · MIN_PRICE_MOVE_PCT 0.00		
KPI Value	N 3		
EDGE +2.58%	P 0.207		
HIT-RATE 66.7%			

KEY FINDINGS

- Detected 14 OI alert signal(s) for BTCUSDT
- Best forward-return signal: WHALE LONG ENTRY +2.58% (n=3, hit-rate 66.7%)

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Generated by the Cerberus studies framework · schema 2

Data Sources

- `duckdb:crypto_binance_oi_data`

Key Findings

- Detected 14 OI alert signal(s) for BTCUSDT
- Best forward-return signal: WHALE LONG ENTRY +2.58% (n=3, hit-rate 66.7%)

Executive Summary

NOTE

Detects leverage build-up and liquidation flush events from open-interest outliers and measures their BTC forward-return impact.

Results & Interpretation

- Across 3 events, the forward return averaged +2.58%. Returns were positive 66.7% of the time. The difference has $p \approx 0.207$.
- Across 1 event, the forward return averaged +1.86%. Returns were positive 100.0% of the time.
- Across 4 events, the forward return averaged -2.94%. Returns were positive 50.0% of the time. The difference has $p \approx 0.557$.
- Across 6 events, the forward return averaged +1.32%. Returns were positive 66.7% of the time. The difference has $p \approx 0.613$.

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Daily open-interest changes are z-scored over a 20-day window. Signals fire when $|z| > 2.0$ and $|\text{price move}| > 0.0008$; forward BTC returns over 7 days are tabulated per signal type with t-tests.

Charts

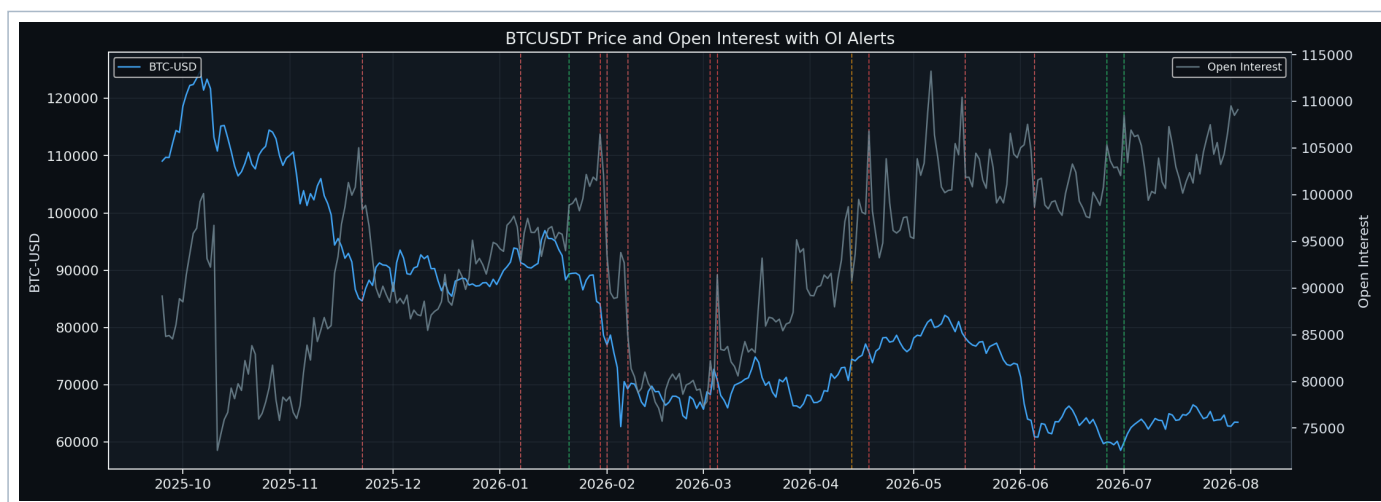


Figure 1. BTCUSDT BTC-USD price and open interest with signals

BTCUSDT BTC-USD price and open interest with signals_

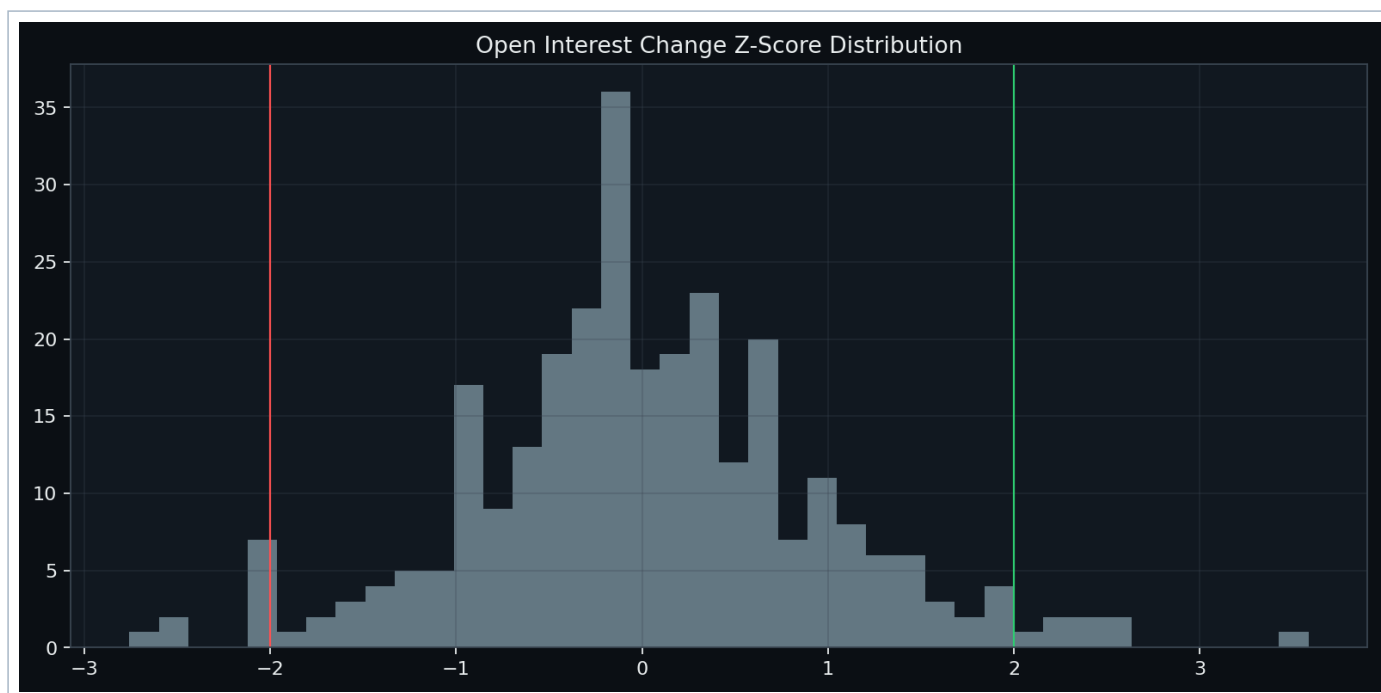


Figure 2. Distribution of 20-day OI change z-scores

Distribution of 20-day OI change z-scores_

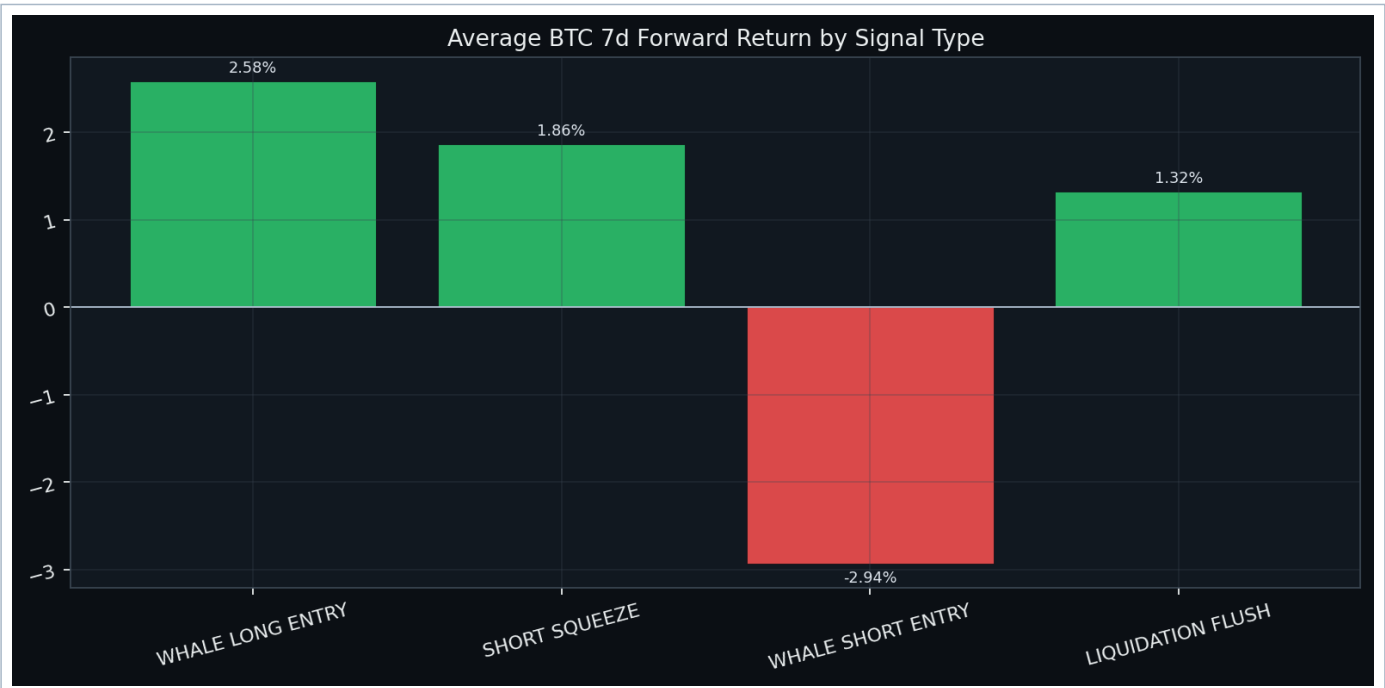


Figure 3. Average BTC 7d forward return by signal type

Average BTC 7d forward return by signal type_

Statistical Results

close	oi	oi_change_...	oi_zscore	price_change_...	fwd_ret_pct
60,922.67	98,688.42	-5.79%	-2.10	-4.51%	+ 4.30%
60,867.41	101,597.51	+ 2.95%	1.05	-0.09%	+ 5.84%
63,239.52	101,776.77	+ 0.18%	0.03	+ 3.90%	+ 3.91%
63,090.59	98,914.16	-2.81%	-0.95	-0.24%	+ 5.07%
61,643.78	98,493.63	-0.43%	-0.07	-2.29%	+ 6.42%
61,449.29	99,232.02	+ 0.75%	0.31	-0.32%	+ 4.83%
63,561.05	99,363.02	+ 0.13%	0.06	+ 3.44%	-1.05%
63,543.20	98,329.49	-1.04%	-0.30	-0.03%	-0.00%
64,421.32	97,795.59	-0.54%	-0.13	+ 1.38%	-0.28%
65,710.40	100,142.27	+ 2.40%	0.97	+ 2.00%	-3.76%
66,289.50	101,679.08	+ 1.53%	0.59	+ 0.88%	-3.53%
65,600.64	103,300.78	+ 1.59%	0.56	-1.04%	-4.47%
64,418.45	102,414.65	-0.86%	-0.40	-1.80%	-5.31%
62,896.47	99,312.69	-3.03%	-1.26	-2.36%	-5.05%
63,540.84	98,609.80	-0.71%	-0.21	+ 1.02%	-5.55%
64,239.68	97,664.21	-0.96%	-0.32	+ 1.10%	-6.69%
63,237.54	97,540.13	-0.13%	0.11	-1.56%	-5.86%

close	oi	oi_change_...	oi_zscore	price_change_...	fwd_ret_pct
63,952.11	100,266.13	+ 2.79%	1.37	+ 1.13%	-5.96%
62,668.02	99,597.37	-0.67%	-0.14	-2.01%	-6.56%
60,995.13	98,871.04	-0.73%	-0.22	-2.67%	-1.63%
59,721.68	100,826.83	+ 1.98%	1.11	-2.09%	+ 2.95%
60,016.43	105,339.46	+ 4.48%	2.33	+ 0.49%	+ 4.21%
59,940.10	103,637.49	-1.62%	-0.92	-0.13%	+ 5.25%
59,532.34	102,912.57	-0.70%	-0.52	-0.68%	+ 6.75%
60,138.38	102,976.53	+ 0.06%	-0.10	+ 1.02%	+ 6.41%
58,558.86	102,060.27	-0.89%	-0.59	-2.63%	+ 8.09%
60,003.76	108,494.20	+ 6.30%	2.61	+ 2.47%	+ 3.76%
61,485.30	103,471.25	-4.63%	-1.97	+ 2.47%	+ 2.78%
62,544.20	106,938.48	+ 3.35%	1.11	+ 1.72%	+ 2.53%
63,088.30	106,236.15	-0.66%	-0.39	+ 0.87%	+ 1.13%
63,547.88	106,397.73	+ 0.15%	-0.04	+ 0.73%	+ 0.33%
63,995.02	105,283.12	-1.05%	-0.46	+ 0.70%	-2.74%
63,297.39	102,690.93	-2.46%	-0.97	-1.09%	+ 2.62%
62,257.57	99,418.28	-3.19%	-1.24	-1.64%	+ 3.94%
63,193.15	100,362.60	+ 0.95%	0.32	+ 1.50%	+ 0.94%
64,127.14	100,146.65	-0.22%	-0.14	+ 1.48%	-0.36%
63,802.03	103,937.50	+ 3.79%	1.27	-0.51%	+ 1.56%
63,758.22	101,386.26	-2.45%	-0.94	-0.07%	+ 1.46%
62,239.12	100,679.52	-0.70%	-0.29	-2.38%	+ 4.81%
64,956.11	107,308.90	+ 6.58%	2.00	+ 4.37%	+ 2.38%
64,712.38	105,378.41	-1.80%	-0.67	-0.38%	+ 2.15%
63,789.28	102,994.61	-2.26%	-0.74	-1.43%	+ 1.97%
63,899.46	101,759.82	-1.20%	-0.39	+ 0.17%	+ 0.31%
64,796.60	100,175.46	-1.56%	-0.49	+ 1.40%	-0.75%
64,690.80	101,424.45	+ 1.25%	0.43	-0.16%	+ 1.00%
65,230.03	102,374.61	+ 0.94%	0.29	+ 0.83%	-2.31%
66,505.12	101,264.63	-1.08%	-0.30	+ 1.95%	-3.96%
66,100.80	104,322.62	+ 3.02%	1.18	-0.61%	-3.32%
65,044.81	102,239.46	-2.00%	-0.74	-1.60%	-0.49%

Research only. Not investment advice. Past performance is not indicative of future results.

close	oi	oi_change_...	oi_zscore	price_change_...	fwd_ret_pct
64,098.50	104,584.91	+ 2.29%	0.95	-1.45%	-2.00%
64,311.81	106,114.82	+ 1.46%	0.58	+ 0.33%	-2.41%
65,340.30	107,509	+ 1.31%	0.47	+ 1.60%	-2.84%
63,724.90	104,354.20	-2.93%	-1.20	-2.47%	-0.41%
63,871.36	105,602.91	+ 1.20%	0.36	+ 0.23%	N/A
63,908.17	103,253.30	-2.22%	-0.97	+ 0.06%	N/A
64,725.31	104,348.71	+ 1.06%	0.33	+ 1.28%	N/A
62,813.75	106,595.94	+ 2.15%	0.84	-2.95%	N/A
62,763.32	109,523.59	+ 2.75%	0.98	-0.08%	N/A
63,482	108,516.03	-0.92%	-0.56	+ 1.15%	N/A
63,460.90	109,126.56	+ 0.56%	0.25	-0.03%	N/A

Table 1. Daily OI/price frame tail

time	type	close	oi	oi_zscore	price_change_...	oi_change_...	fwd_ret_pct
2025-11-22	LIQUIDATION FLUSH	84,648.36	98,408.86	-2.44	-0.52%	-6.32%	+ 7.33%
2026-01-07	LIQUIDATION FLUSH	91,308.05	92,969.52	-2.04	-2.58%	-3.70%	+ 6.16%
2026-01-21	WHALE LONG ENTRY	89,376.96	98,893.90	2.30	+ 1.21%	+ 5.19%	-0.22%
2026-01-30	WHALE SHORT ENTRY	84,128.66	106,511.19	2.12	-0.51%	+ 4.90%	-16.13%
2026-02-01	LIQUIDATION FLUSH	76,974.45	93,551.30	-2.76	-2.09%	-8.43%	-8.72%
2026-02-07	LIQUIDATION FLUSH	69,281.97	85,124.11	-2.05	-1.80%	-8.19%	+ 0.70%
2026-03-03	WHALE SHORT ENTRY	68,293.65	82,214.75	2.56	-0.70%	+ 5.63%	+ 2.39%
2026-03-05	WHALE SHORT ENTRY	70,841.12	91,427.87	3.58	-2.57%	+ 15.53%	-0.49%
2026-04-13	SHORT SQUEEZE	74,484.64	90,771.45	-2.45	+ 5.27%	-8.06%	+ 1.86%
2026-04-18	WHALE SHORT ENTRY	75,726.21	106,822.36	2.23	-1.82%	+ 9.07%	+ 2.49%
2026-05-16	LIQUIDATION FLUSH	78,131.44	101,877.65	-2.03	-1.18%	-7.76%	-1.87%
2026-06-05	LIQUIDATION FLUSH	60,922.67	98,688.42	-2.10	-4.51%	-5.79%	+ 4.30%
2026-06-26	WHALE LONG ENTRY	60,016.43	105,339.46	2.33	+ 0.49%	+ 4.48%	+ 4.21%
2026-07-01	WHALE LONG ENTRY	60,003.76	108,494.20	2.61	+ 2.47%	+ 6.30%	+ 3.76%

Table 2. Detected OI alert signals

signal	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_...	t_stat	p_value
WHALE LONG ENTRY	3	+ 2.58%	+ 3.76%	+ 66.67%	1.84	- 0.207
SHORT SQUEEZE	1	+ 1.86%	+ 1.86%	+ 100.00%	N/A	N/A
WHALE SHORT ENTRY	4	-2.94%	+ 0.95%	+ 50.00%	-0.66	- 0.557
LIQUIDATION FLUSH	6	+ 1.32%	+ 2.50%	+ 66.67%	0.54	- 0.613

Table 3. Forward-return statistics by signal type

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Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

frame

date	close	oi	oi_change_...	oi_zscore	price_change_...	fwd_ret_pct
2026-06-05	60,922.67	98,688.42	-5.79%	-2.10	-4.51%	+ 4.30%
2026-06-06	60,867.41	101,597.51	+ 2.95%	1.05	-0.09%	+ 5.84%
2026-06-07	63,239.52	101,776.77	+ 0.18%	0.03	+ 3.90%	+ 3.91%
2026-06-08	63,090.59	98,914.16	-2.81%	-0.95	-0.24%	+ 5.07%
2026-06-09	61,643.78	98,493.63	-0.43%	-0.07	-2.29%	+ 6.42%
2026-06-10	61,449.29	99,232.02	+ 0.75%	0.31	-0.32%	+ 4.83%
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2026-06-14	65,710.40	100,142.27	+ 2.40%	0.97	+ 2.00%	-3.76%
2026-06-15	66,289.50	101,679.08	+ 1.53%	0.59	+ 0.88%	-3.53%
2026-06-16	65,600.64	103,300.78	+ 1.59%	0.56	-1.04%	-4.47%
2026-06-17	64,418.45	102,414.65	-0.86%	-0.40	-1.80%	-5.31%
2026-06-18	62,896.47	99,312.69	-3.03%	-1.26	-2.36%	-5.05%
2026-06-19	63,540.84	98,609.80	-0.71%	-0.21	+ 1.02%	-5.55%

signal_stats

signal	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_...	t_stat	p_value
WHALE LONG ENTRY	3	+ 2.58%	+ 3.76%	+ 66.67%	1.84	- 0.207
SHORT SQUEEZE	1	+ 1.86%	+ 1.86%	+ 100.00%	N/A	N/A
WHALE SHORT ENTRY	4	-2.94%	+ 0.95%	+ 50.00%	-0.66	- 0.557
LIQUIDATION FLUSH	6	+ 1.32%	+ 2.50%	+ 66.67%	0.54	- 0.613

signals

time	type	close	oi	oi_zscore	price_change_...	oi_change_...	fwd_ret_pct
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Statistics Summary

Key	Value
study	btc_oi_alerts
category	crypto
status	success
rows	313
n_signals	14
latest_oi_z	0.25
horizon	7
symbol	BTCUSDT
git_commit	aa4576bdd6426fdeffd63347e0768e52b6c94632
executed_at	2026-08-05T08:29:10.629355+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-05 08:29 UTC
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Study slug	btc_oi_alerts
Run ID	20260805_092910

Field	Value
Git commit	aa4576bd
Datasets	duckdb:crypto_binance_oi_data
Parameters hash	9c86c766c2cc
Parameter · symbol	BTCUSDT
Parameter · benchmark	BTC-USD
Parameter · lookback	20
Parameter · horizon	7
Parameter · z_threshold	2
Parameter · min_price_move_pct	0.00
Data fingerprint	bc6071cb21f0
Duration	0.5s

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